

Change CAmkES examples from sel4-tutorial-manifest

- [Get examples and install the sources](#)
 - [Prepare your system for the installation](#)
 - [Compile and run](#)
 - [Initialize](#)
 - [Compile](#)
 - [Modify](#)
 - [open a *.md file and looking for the C code](#)
-

Get examples and install the sources

The official tutorials for CAmkES will be find in <https://docs.sel4.systems/Tutorials> .

Prepare your system for the installation

Make sure that **repo** is already installed. Additional python dependencies are required to build tutorials.

```
pip install --user aenum pip install --user pyelftools
```

You can install the examples with the following statements. If you have already installed pip on your system, you only need to start an update. Execute the following statement: `pip install --upgrade pip`

```
mkdir <DIR_NAME>
cd <DIR_NAME>
repo init -u https://github.com/SEL4PROJ/sel4-tutorials-manifest
repo sync
```

NOTE: Placeholder **<DIR_NAME>** is representative for fully qualified path name

Compile and run

All examples are located in path `<DIR_PATH>/projects/sel4-tutorials/tutorials` from your installation.

Before start building run the docker container in QEMU with instruction

```
make -C <DIR_PATH>sel4-CAmkES-L4v-dockerfiles user_camkes HOST_DIR=$(pwd)
```

from the path where you can find your installation directory and afterwards change to the directory with your installation.

If you want to compile any projects follow these steps.

Initialize

execute the statement **init** in your installation directory with specified parameter:

```
--tut <PROJECT_NAME>
```

HINT: The **<PROJECT_NAME>** has the same identifier like your created project folder

If you deal with other identifiers, like the project directories identifiers, you become an error message.

```
--plat <PLATFORM>
```

HINT: The current **<PLATFORM>** is `zynq7000`

This script create two directories in your installation directory with the identifier from your given project name. One directory name is identical with the given project name, the other contain the extension **_build**. You can find the source code in the directory with the identical identifier. The directory with the extension **_build** can be compiled and then executed.

Compile

change to the directory with the extension **_build** and compile this project with **ninja** and run with the script **simulate** your project will be running

If you want to change, to adjust or to expand the sources, so you have to make some changes in the files from the installation. It is possible to have a look to the directory with the identical identifier, but changes doesn't have influence your **_build** directory. So you have to compile and run the project in a new way. To do this, make changes and follow the descriptive steps (step 1 and 2) again.

Modify

have a look to the directory <DIR_PATH>/projects/sel4-tutorials/tutorials and choose your project which you want to be changed

Enter to a project directory and you can find a file names with the extension **.md** . script explained in the beginning, which is responsible for compiling the project invokes functions from Python scripts, which you can find in path <DIR_PATH>/projects/sel4-tutorials and his subdirectory /tool . The content from the file with the extension **.md** will be parsed and additional rendered and then generated by the Python script to the source file in the created project folders.

So source code changes will be doing at the **.md** file.

open a *.md file and looking for the C code

change or complete everything in this file and retry the sub steps one and two

HINT: It is better if you remove the previously created project folders .